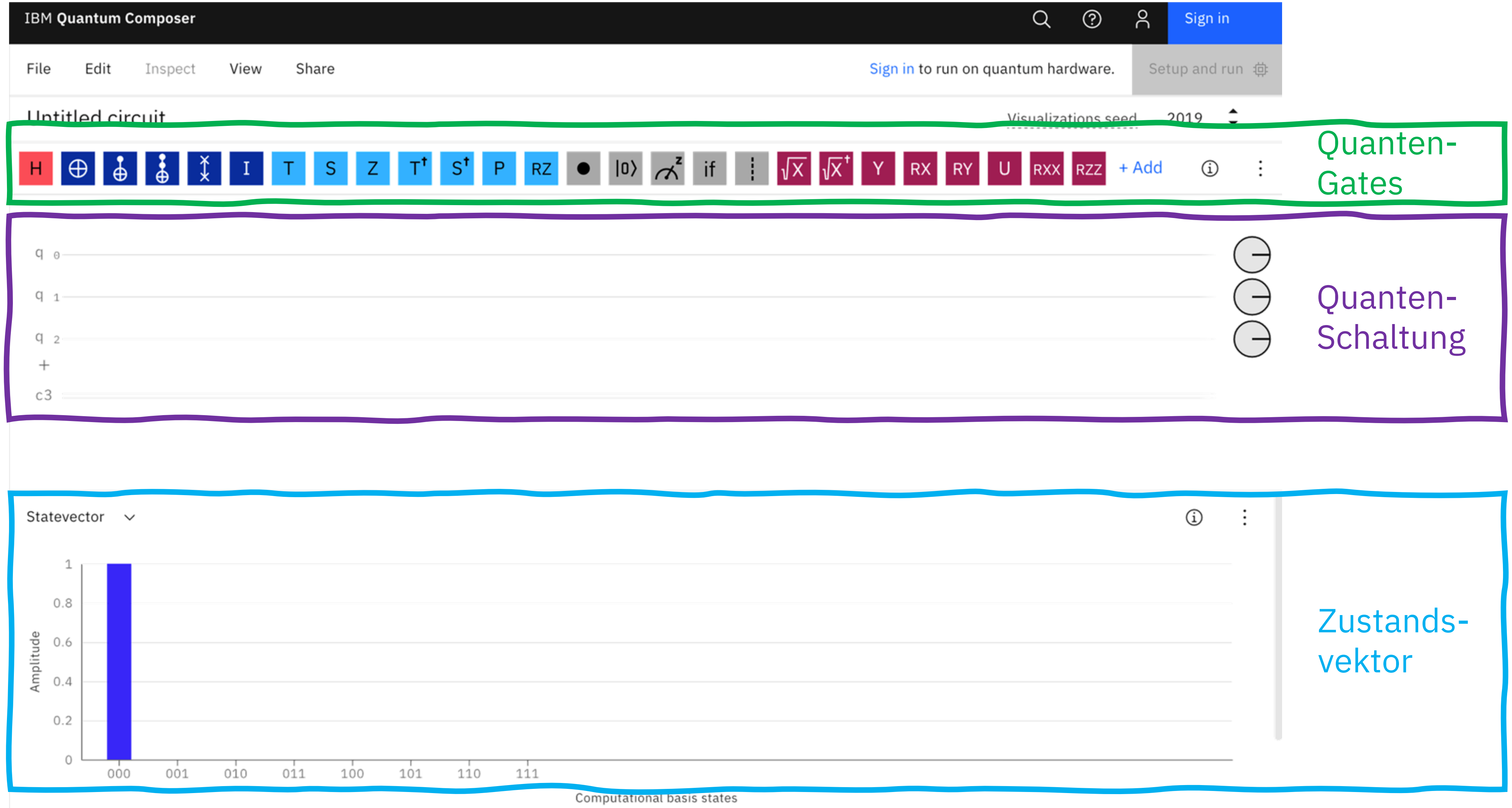
IBM Quantum

Was ist der IBM Quantum Composer?

Isabel Nha Minh Le
IBM Quantum Community

IBM **Quantum**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File
Edit
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View
Share
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Setup and run
Untitled circuit
Visualizations seed
2019
H
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S
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P
RZ
if
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RX
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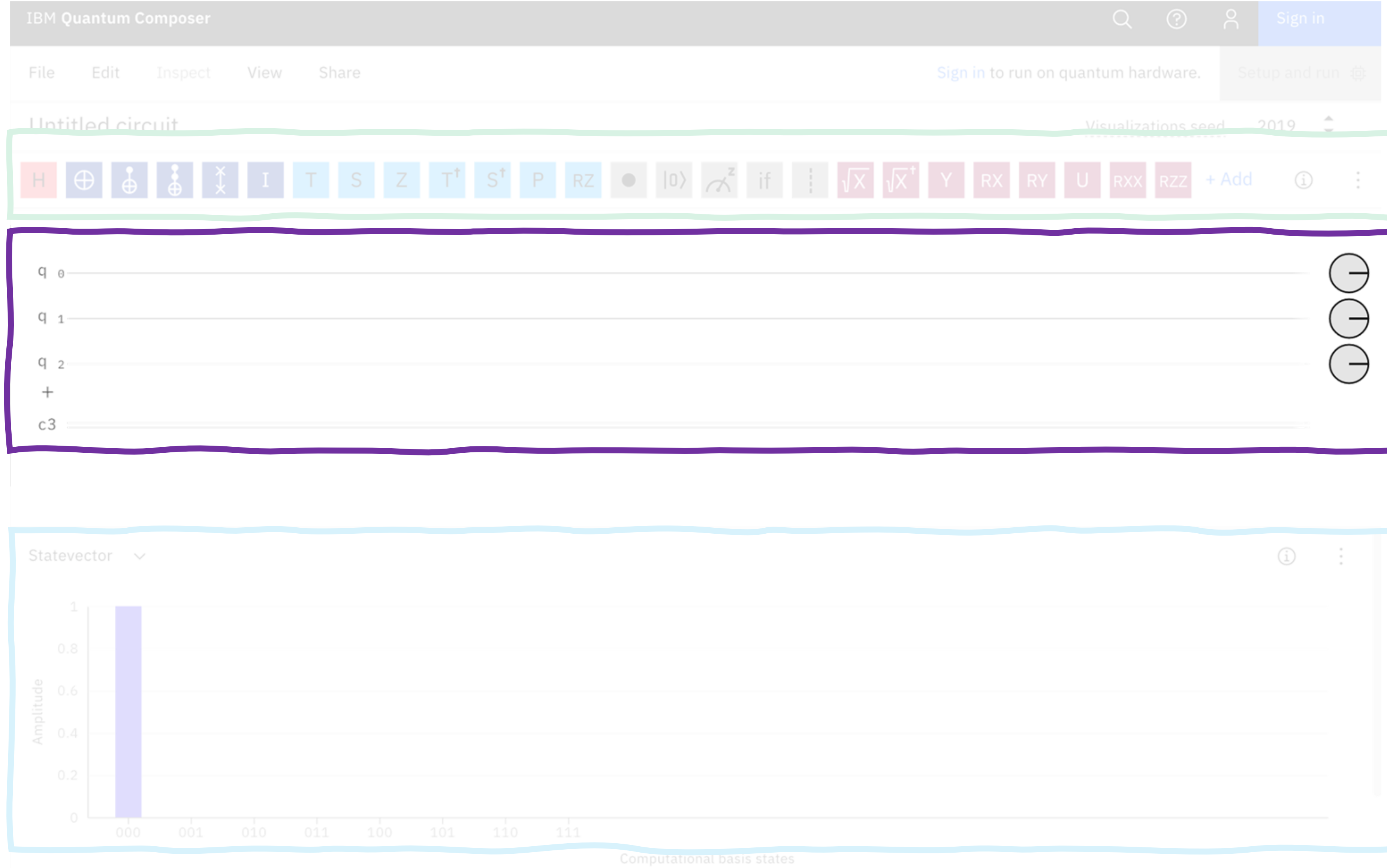
RZZ
+ Add
Quanten-Gates
Quanten-Schaltung
c3
Statevector
Amplitude
Computational basis states
Zustands-vektor



Quanten-Gates

Quanten-Schaltung

Zustands-vektor

Quanten-Schaltung

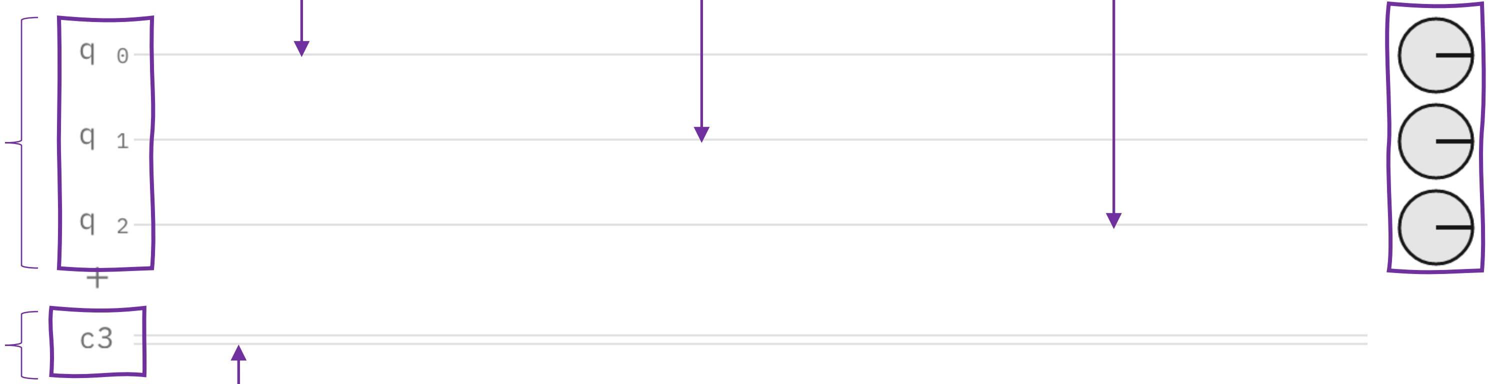
0. Qubit

1. Qubit

2. Qubit

3 Qubits

3 Bits

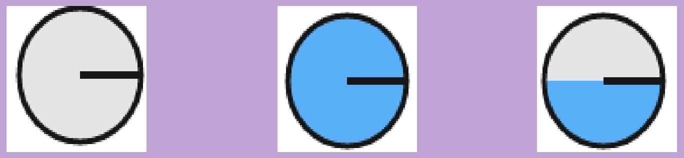


Visualisierung des *Qubit-Zustands*

„Superposition“

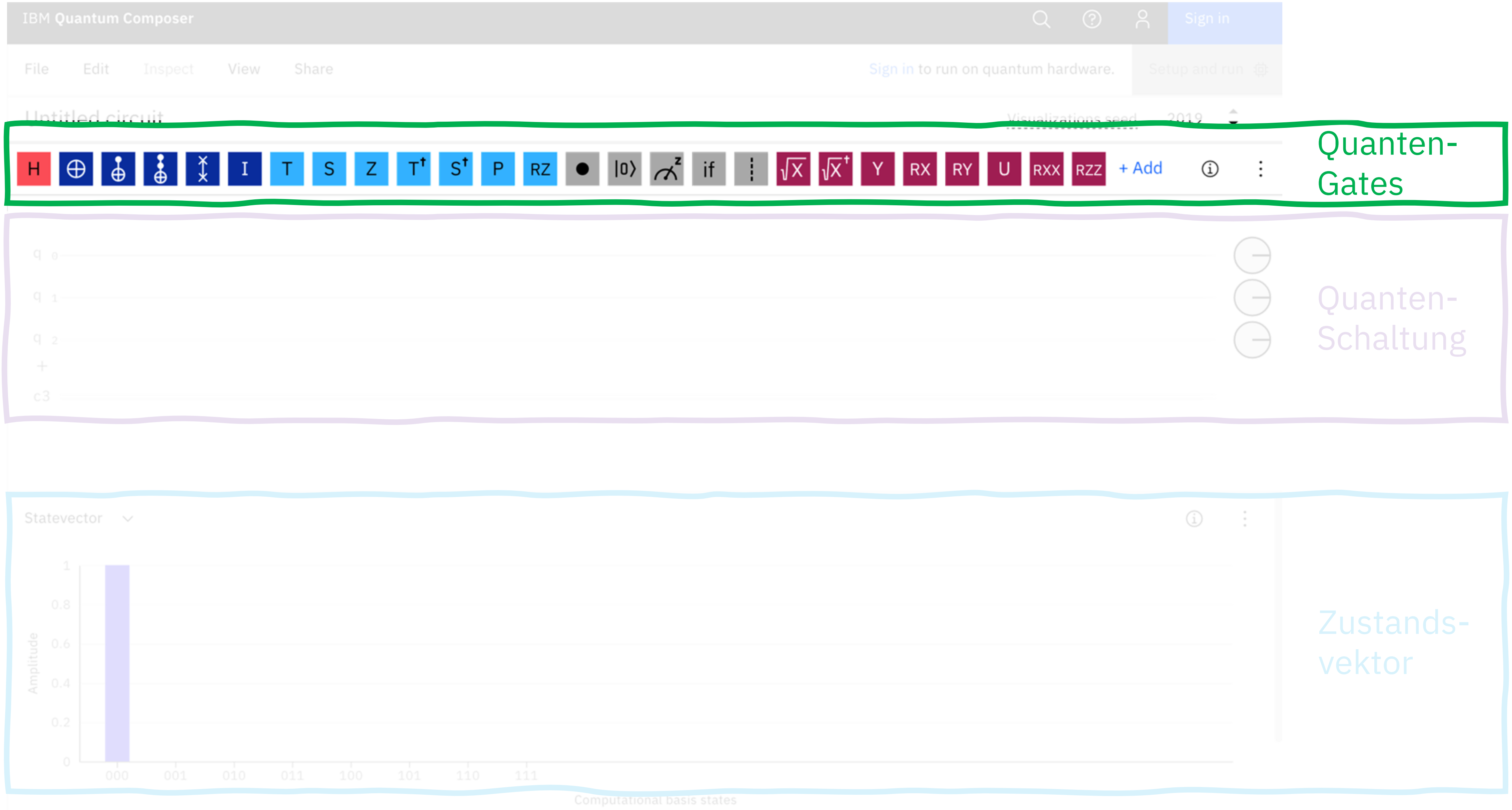
$\alpha|0\rangle + \beta|1\rangle$

Das Ergebnis der Messung eines Qubits wird als klassische Information (also 0 oder 1) in einem Bit gespeichert



$|0\rangle$ $|1\rangle$ $\frac{|0\rangle + |1\rangle}{\sqrt{2}}$

IBM **Quantum**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S
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P
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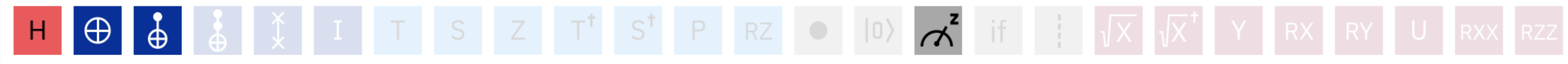
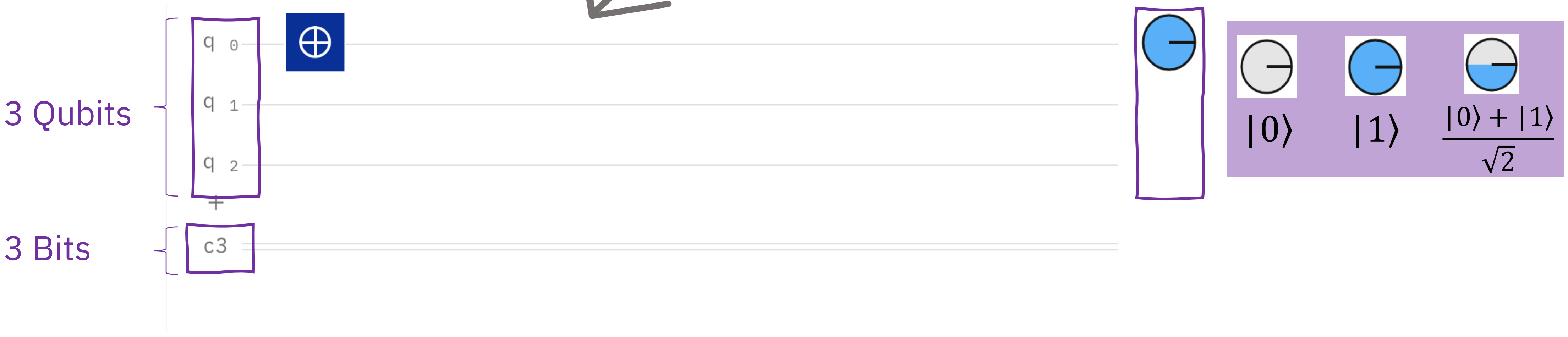
000
001
010
011
100
101
110
111
Computational basis states
Zustands-vektor

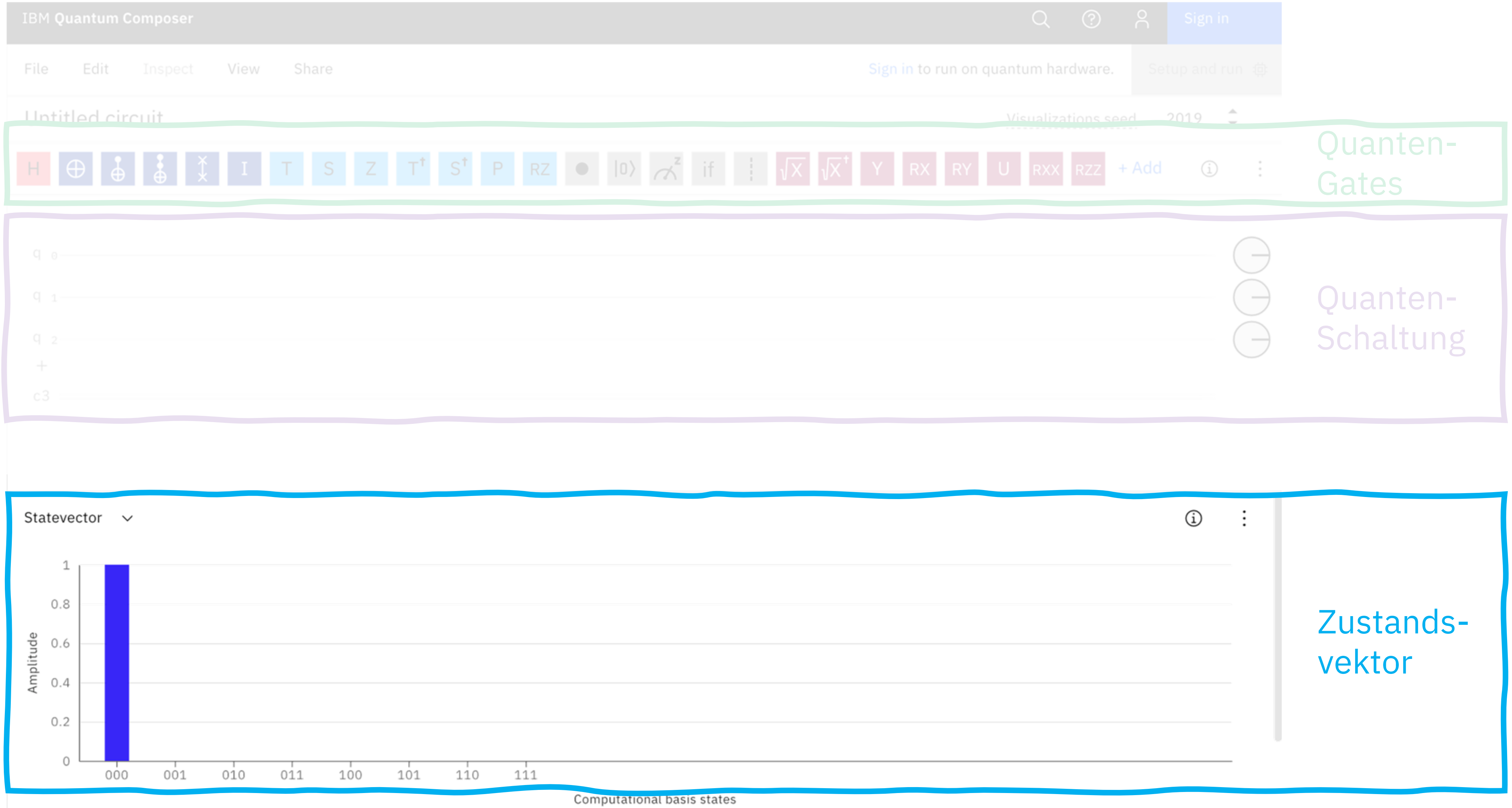
Quanten-Gates

- Quanten-Gates verändern den Zustand eines Qubits



Quanten-Schaltung

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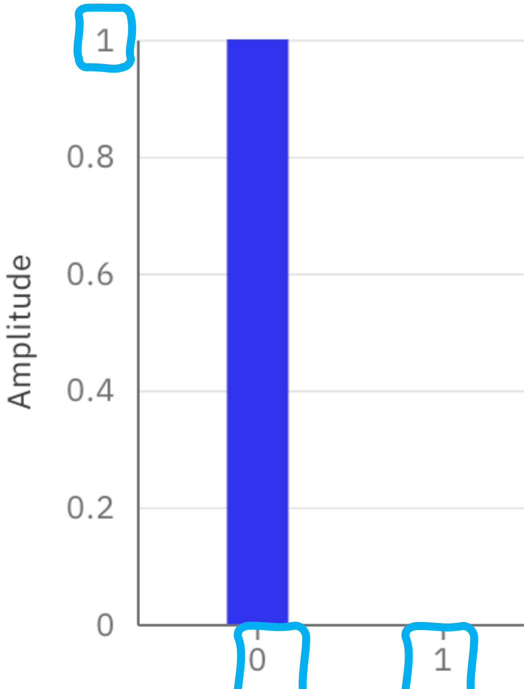
+ Add
Quanten-Gates
q 0
q 1
q 2
c3
Quanten-Schaltung
Statevector
Amplitude
1
0.8
0.6
0.4
0.2
0
000 001 010 011 100 101 110 111
Computational basis states
Zustands-vektor

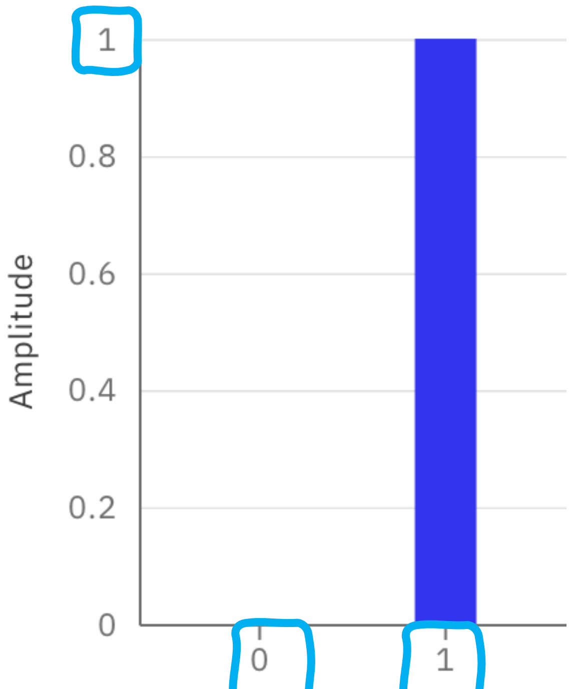
Zustandsvektor (Amplituden)

1 Qubit: $\alpha|0\rangle + \beta|1\rangle$

Amplituden



$\alpha|0\rangle + \beta|1\rangle$
$= 1 \cdot |0\rangle$



$\alpha|0\rangle + \beta|1\rangle$
$= 1 \cdot |1\rangle$

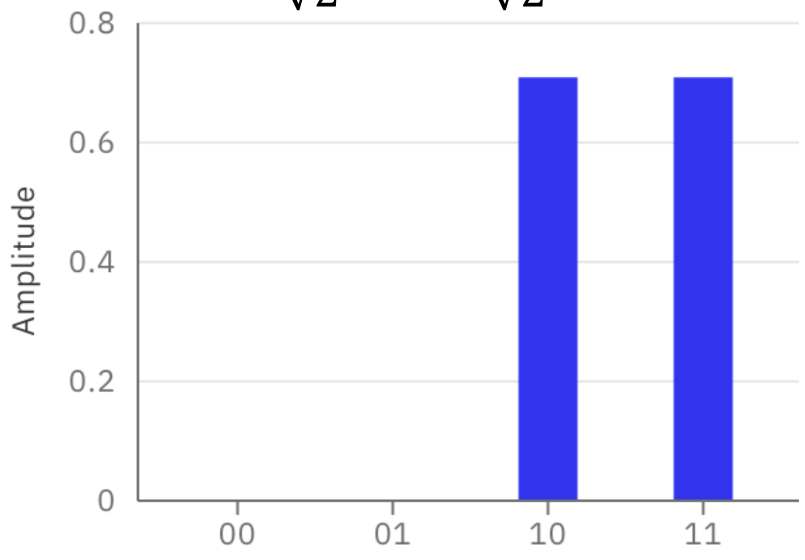
Zustandsvektor (Amplituden)

1 Qubit: $\alpha|0\rangle + \beta|1\rangle$

2 Qubits: $\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$

3 Qubits: $\alpha|000\rangle + \beta|001\rangle + \gamma|010\rangle + \delta|011\rangle + \varepsilon|100\rangle + \theta|101\rangle + \vartheta|110\rangle + \mu|111\rangle$

$$\frac{1}{\sqrt{2}}|10\rangle + \frac{1}{\sqrt{2}}|11\rangle$$



$$\frac{1}{\sqrt{2}}|010\rangle + \frac{1}{\sqrt{2}}|101\rangle$$